



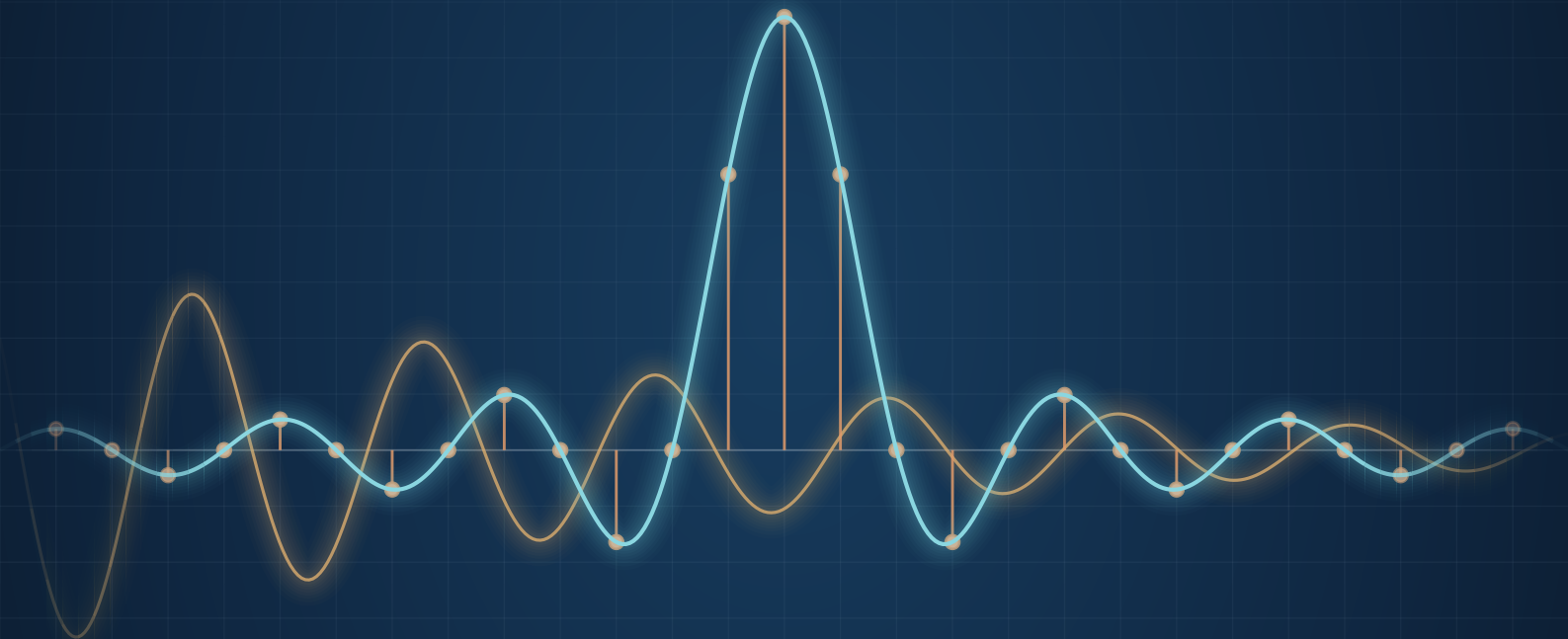
SIGNALS AND SYSTEMS

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# Signals, Systems and Frequency-Domain Analysis

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*Formula and Notation Reference*



# Contents

*The conventions used throughout the course, every formula it establishes, and every symbol it defines. Nothing here is derived; the derivations are in the lecture notes.*

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## How to read it

Part 1 states the conventions and the two transform pairs. Part 2 is the summary of formulas, in the order the course establishes them. Part 3 defines every symbol. Nothing here is derived: where a result needs an argument, the argument is in the lecture notes chapter named beside it.

## PART

## 1

# Conventions

*These hold everywhere in the course, without local variation. Where a result depends on one of them, it is restated at that point rather than assumed.*

| CONVENTION        | STATEMENT                                                                                                                                          |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Energy and power  | Normalised: the resistance is taken as $1 \Omega$ , so instantaneous power is $ x ^2$ and energy is its integral or sum.                           |
| Imaginary unit    | $j$ , with $j^2 = -1$ .                                                                                                                            |
| Angular frequency | $\omega$ in rad/s in continuous time and rad/sample in discrete time. A frequency in hertz is written out as such, and $\omega = 2\pi f$ .         |
| Brackets          | Round brackets for continuous time, $x(t)$ . Square brackets for discrete time, $x[n]$ , where $n$ is an integer.                                  |
| Convolution       | $*$ . It applies only to systems that are both linear and time invariant.                                                                          |
| sinc              | Unnormalised, $\text{sinc}(\theta) = \sin \theta / \theta$ , with zeros at $\theta = \pm k\pi$ . The convention is restated wherever sinc is used. |
| Sampling          | $\omega_s = 2\pi/T$ rad/s and $f_s = 1/T$ Hz. The sampling theorem is stated with a strict inequality, $\omega_s > 2\omega_M$ .                    |

## THE CONTINUOUS-TIME FOURIER TRANSFORM PAIR

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega)e^{j\omega t} d\omega$$

The first is the analysis equation and the second the synthesis equation. The factor  $\frac{1}{2\pi}$  belongs to synthesis.

## THE DISCRETE-TIME FOURIER TRANSFORM PAIR

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$

$$x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega})e^{j\omega n} d\omega$$

The synthesis integral runs over one period, because the spectrum repeats every  $2\pi$ .

PART

# 2

## Summary of formulas

Everything the course establishes, in the order it establishes it.

### A.1 Energy and power

Energy and power in continuous and discrete time.

| QUANTITY      | CONTINUOUS TIME                                                               | DISCRETE TIME                                                                  |
|---------------|-------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Total energy  | $E_\infty = \int_{-\infty}^{\infty}  x(t) ^2 dt$                              | $E_\infty = \sum_{n=-\infty}^{\infty}  x[n] ^2$                                |
| Average power | $P_\infty = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T  x(t) ^2 dt$ | $P_\infty = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N  x[n] ^2$ |
| Energy signal | $E_\infty < \infty, P_\infty = 0$                                             | same                                                                           |
| Power signal  | $E_\infty \rightarrow \infty, 0 < P_\infty < \infty$                          | same                                                                           |

### A.2 Operations and periodicity

Operations on the time axis, and the test for periodicity.

| ITEM                  | STATEMENT                                                                                                                  |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------|
| Shift                 | $x(t - t_0)$ : delay if $t_0 > 0$ , advance if $t_0 < 0$                                                                   |
| Reversal              | $x(-t), x[-n]$                                                                                                             |
| Scaling               | $y(t) = x(at)$ : compressed if $a > 1$ , stretched if $0 < a < 1$ ; support divided by $a$                                 |
| Combination           | $x(at - b)$ : shift by $b$ first, then scale by $a$                                                                        |
| Periodicity           | $x(t) = x(t + T); x[n] = x[n + N]$ with $N$ an integer                                                                     |
| Fundamental frequency | $\omega_0 = 2\pi/T_0 = 2\pi/N_0$                                                                                           |
| Even and odd          | $\mathcal{E}\mathbf{v}\{x\} = \frac{1}{2}[x(t) + x(-t)], \mathcal{O}\mathbf{d}\mathbf{d}\{x\} = \frac{1}{2}[x(t) - x(-t)]$ |

## A.3 Impulses and steps

The unit impulse and the unit step in continuous and discrete time.

| ITEM             | CONTINUOUS TIME                                                            | DISCRETE TIME                                                       |
|------------------|----------------------------------------------------------------------------|---------------------------------------------------------------------|
| Step and impulse | $\delta(t) = \frac{d}{dt}u(t), u(t) = \int_{-\infty}^t \delta(\tau) d\tau$ | $\delta[n] = u[n] - u[n-1], u[n] = \sum_{k=0}^{\infty} \delta[n-k]$ |
| Sampling         | $x(t)\delta(t-t_0) = x(t_0)\delta(t-t_0)$                                  | $x[n]\delta[n-n_0] = x[n_0]\delta[n-n_0]$                           |
| Sifting          | $x(t_0) = \int_{-\infty}^{\infty} x(t)\delta(t-t_0) dt$                    | $x[n_0] = \sum_n x[n]\delta[n-n_0]$                                 |
| Representation   | $x(t) = \int x(\tau)\delta(t-\tau) d\tau$                                  | $x[n] = \sum_k x[k]\delta[n-k]$                                     |

## A.4 Complex exponentials

Complex exponentials in continuous and discrete time, and their periods.

| ITEM                  | STATEMENT                                                                                                       |
|-----------------------|-----------------------------------------------------------------------------------------------------------------|
| Continuous time       | $x(t) = Ce^{at}$ , with $C = Ae^{j\theta}$ and $a = r + j\omega_0$ : $x(t) = Ae^{rt}e^{j(\omega_0 t + \theta)}$ |
| Period                | $T_0 = 2\pi/\omega_0$ , always periodic for $\omega_0 \neq 0$                                                   |
| Discrete time         | $x[n] = C\alpha^n$ with $\alpha = e^{\beta}$ ; growth boundary at $ \alpha  = 1$                                |
| Period                | $N = \frac{2\pi}{\omega_0}k$ ; periodic only if $\omega_0/2\pi \in \mathbb{Q}$                                  |
| Frequency wrap-around | $e^{j(\omega_0+2\pi)n} = e^{j\omega_0 n}$                                                                       |

## A.5 Systems and convolution

System properties, in general and for an LTI system in terms of  $h$ .

| PROPERTY       | GENERAL CRITERION                     | LTI CRITERION IN TERMS OF $h$                     |
|----------------|---------------------------------------|---------------------------------------------------|
| Memoryless     | output at $t$ uses only input at $t$  | $h(t) = a\delta(t)$ or $h[n] = a\delta[n]$        |
| Invertible     | distinct inputs give distinct outputs | $h * g = \delta$ for some $g$                     |
| Causal         | output uses only $\tau \leq t$        | $h(t) = 0$ for $t < 0$ ; $h[n] = 0$ for $n < 0$   |
| BIBO stable    | bounded input gives bounded output    | $\int  h  dt < \infty$ ; $\sum_k  h[k]  < \infty$ |
| Time invariant | $x(t-t_0) \rightarrow y(t-t_0)$       | —                                                 |
| Linear         | $ax_1 + bx_2 \rightarrow ay_1 + by_2$ | —                                                 |

## CONVOLUTION

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau) \, d\tau$$

Convolution is commutative, distributive, and associative. The output support is the sum of the input supports. The total areas, or the total sums in discrete time, multiply.

## A.7 Fourier series properties

A periodic signal of period  $T_0$  with  $\omega_0 = 2\pi/T_0$ , or a periodic sequence of period  $N$  with  $\omega_0 = 2\pi/N$ . In both columns  $x \leftrightarrow a_k$  and  $y \leftrightarrow b_k$ .

Fourier series properties in continuous and discrete time.

| PROPERTY                 | CONTINUOUS TIME                                                                                                             | DISCRETE TIME                                                                                       |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Linearity                | $Ax(t) + By(t) \leftrightarrow Aa_k + Bb_k$                                                                                 | $Ax[n] + By[n] \leftrightarrow Aa_k + Bb_k$                                                         |
| Time shift               | $x(t - t_0) \leftrightarrow a_k e^{-jk\omega_0 t_0}$                                                                        | $x[n - n_0] \leftrightarrow a_k e^{-jk(2\pi/N)n_0}$                                                 |
| Frequency shift          | $e^{jM\omega_0 t} x(t) \leftrightarrow a_{k-M}$                                                                             | $e^{jM(2\pi/N)n} x[n] \leftrightarrow a_{k-M}$                                                      |
| Conjugation              | $x^*(t) \leftrightarrow a_{-k}^*$                                                                                           | $x^*[n] \leftrightarrow a_{-k}^*$                                                                   |
| Time reversal            | $x(-t) \leftrightarrow a_{-k}$                                                                                              | $x[-n] \leftrightarrow a_{-k}$                                                                      |
| Time scaling             | $x(\alpha t), \alpha > 0: a_k, \text{ period } T_0/\alpha$                                                                  | $x_{(m)}[n]: \frac{1}{m}a_k, \text{ period } mN$                                                    |
| Periodic convolution     | $\int_{T_0} x(\tau)y(t - \tau)d\tau \leftrightarrow T_0 a_k b_k$                                                            | $\sum_{r=\langle N \rangle} x[r]y[n - r] \leftrightarrow N a_k b_k$                                 |
| Multiplication           | $x(t)y(t) \leftrightarrow \sum_{\ell=-\infty}^{\infty} a_\ell b_{k-\ell}$                                                   | $x[n]y[n] \leftrightarrow \sum_{\ell=\langle N \rangle} a_\ell b_{k-\ell}$                          |
| Derivative or difference | $dx/dt \leftrightarrow jk\omega_0 a_k$                                                                                      | $x[n] - x[n - 1] \leftrightarrow (1 - e^{-jk(2\pi/N)})a_k$                                          |
| Integral or running sum  | $\int_{-\infty}^t x d\tau \leftrightarrow \frac{a_k}{jk\omega_0}, \text{ only if } a_0 = 0$                                 | $\sum_{r=-\infty}^n x[r] \leftrightarrow \frac{a_k}{1 - e^{-jk(2\pi/N)}}, \text{ only if } a_0 = 0$ |
| Real signal              | $a_k = a_{-k}^*;  a_k  \text{ even, } \angle a_k \text{ odd}$                                                               | same                                                                                                |
| Real and even            | $a_k$ real and even                                                                                                         | same                                                                                                |
| Real and odd             | $a_k$ purely imaginary and odd                                                                                              | same                                                                                                |
| Even-odd parts           | $\mathcal{E}\mathbf{v}\{x\} \leftrightarrow \text{Re}\{a_k\}, \mathcal{O}\text{dd}\{x\} \leftrightarrow j \text{Im}\{a_k\}$ | same                                                                                                |
| Parseval                 | $\frac{1}{T_0} \int_{T_0}  x ^2 dt = \sum_k  a_k ^2$                                                                        | $\frac{1}{N} \sum_{n=\langle N \rangle}  x ^2 = \sum_{k=\langle N \rangle}  a_k ^2$                 |

The two columns differ in three rows because discrete-time coefficients repeat,  $a_k = a_{k+N}$ . For multiplication, the discrete-time sum therefore covers one period instead of all integers. Time expansion carries the factor  $\frac{1}{m}$  because the average uses  $m$  times as many samples. Discrete time also uses a first difference where continuous time uses a derivative. At  $k = 0$ , integration or accumulation requires zero mean; otherwise the result grows without bound.

## A.8 Continuous-time Fourier transform

Properties of the continuous-time Fourier transform.

| PROPERTY                     | STATEMENT                                                                                                                          |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Linearity                    | $ax_1(t) + bx_2(t) \leftrightarrow aX_1(j\omega) + bX_2(j\omega)$                                                                  |
| Time shift                   | $x(t - t_0) \leftrightarrow e^{-j\omega t_0} X(j\omega)$                                                                           |
| Frequency shift              | $e^{j\omega_0 t} x(t) \leftrightarrow X(j(\omega - \omega_0))$                                                                     |
| Conjugation                  | $x^*(t) \leftrightarrow X^*(-j\omega)$                                                                                             |
| Time reversal                | $x(-t) \leftrightarrow X(-j\omega)$                                                                                                |
| Time and frequency scaling   | $x(at) \leftrightarrow \frac{1}{ a } X(j\omega/a)$                                                                                 |
| Convolution                  | $x(t) * h(t) \leftrightarrow X(j\omega)H(j\omega)$                                                                                 |
| Multiplication               | $x(t)y(t) \leftrightarrow \frac{1}{2\pi} X(j\omega) * Y(j\omega)$                                                                  |
| Differentiation in time      | $\frac{d^n x}{dt^n} \leftrightarrow (j\omega)^n X(j\omega)$                                                                        |
| Integration                  | $\int_{-\infty}^t x(\tau) d\tau \leftrightarrow \frac{1}{j\omega} X(j\omega) + \pi X(0)\delta(\omega)$                             |
| Differentiation in frequency | $t x(t) \leftrightarrow j \frac{dX(j\omega)}{d\omega}$                                                                             |
| Duality                      | $X(t) \leftrightarrow 2\pi x(-\omega)$                                                                                             |
| Real signal                  | $X(-j\omega) = X^*(j\omega);  X  \text{ even, } \angle X \text{ odd}$                                                              |
| Real and even                | $X(j\omega) \text{ real and even}$                                                                                                 |
| Real and odd                 | $X(j\omega) \text{ purely imaginary and odd}$                                                                                      |
| Even-odd parts               | $\mathcal{E}\mathbf{v}\{x\} \leftrightarrow \text{Re}\{X\}, \mathcal{O}\mathbf{d}\mathbf{d}\{x\} \leftrightarrow j \text{Im}\{X\}$ |
| Parseval                     | $\int_{-\infty}^{\infty}  x(t) ^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty}  X(j\omega) ^2 d\omega$                              |

Continuous-time Fourier transform pairs.

| SIGNAL                                                    | TRANSFORM                                                                         |
|-----------------------------------------------------------|-----------------------------------------------------------------------------------|
| $\delta(t)$                                               | 1                                                                                 |
| $\delta(t - t_0)$                                         | $e^{-j\omega t_0}$                                                                |
| $u(t)$                                                    | $\frac{1}{j\omega} + \pi\delta(\omega)$                                           |
| $e^{-at}u(t), a > 0$                                      | $\frac{1}{a+j\omega}$                                                             |
| $te^{-at}u(t), a > 0$                                     | $\frac{1}{(a+j\omega)^2}$                                                         |
| $\frac{t^{n-1}}{(n-1)!}e^{-at}u(t), a > 0$                | $\frac{1}{(a+j\omega)^n}$                                                         |
| $e^{-a t }, a > 0$                                        | $\frac{2a}{a^2+\omega^2}$                                                         |
| 1 on $ t  < T_1$ , 0 beyond                               | $\frac{2\sin(\omega T_1)}{\omega} = 2T_1 \text{sinc}(\omega T_1)$                 |
| $\frac{\sin(Wt)}{\pi t} = \frac{W}{\pi} \text{sinc}(Wt)$  | 1 on $ \omega  < W$ , 0 beyond                                                    |
| 1                                                         | $2\pi\delta(\omega)$                                                              |
| $e^{j\omega_0 t}$                                         | $2\pi\delta(\omega - \omega_0)$                                                   |
| $\cos \omega_0 t$                                         | $\pi\delta(\omega - \omega_0) + \pi\delta(\omega + \omega_0)$                     |
| $\sin \omega_0 t$                                         | $\frac{\pi}{j}\delta(\omega - \omega_0) - \frac{\pi}{j}\delta(\omega + \omega_0)$ |
| $\sum_k a_k e^{jk\omega_0 t}$                             | $\sum_k 2\pi a_k \delta(\omega - k\omega_0)$                                      |
| Square wave: 1 on $ t  < T_1$ , 0 on $T_1 <  t  \leq T/2$ | $\sum_k \frac{2\sin(k\omega_0 T_1)}{k} \delta(\omega - k\omega_0)$                |
| $\sum_k \delta(t - kT)$                                   | $\frac{2\pi}{T} \sum_k \delta(\omega - \frac{2\pi k}{T})$                         |

This table uses the unnormalised definition  $\text{sinc}(\theta) = \sin \theta / \theta$ . The normalised convention divides the argument by  $\pi$ . Convert the argument when moving between conventions; otherwise the result loses a factor of  $\pi$ . The exponential pairs require  $a > 0$ .

## A.9 Discrete-time Fourier transform

Properties of the discrete-time Fourier transform.

| PROPERTY                     | STATEMENT                                                                                                                        |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Linearity                    | $ax_1[n] + bx_2[n] \leftrightarrow aX_1(e^{j\omega}) + bX_2(e^{j\omega})$                                                        |
| Time shift                   | $x[n - n_0] \leftrightarrow e^{-j\omega n_0} X(e^{j\omega})$                                                                     |
| Frequency shift              | $e^{j\omega_0 n} x[n] \leftrightarrow X(e^{j(\omega - \omega_0)})$                                                               |
| Conjugation                  | $x^*[n] \leftrightarrow X^*(e^{-j\omega})$                                                                                       |
| Time reversal                | $x[-n] \leftrightarrow X(e^{-j\omega})$                                                                                          |
| Time expansion               | $x_{(k)}[n] \leftrightarrow X(e^{jk\omega})$                                                                                     |
| Convolution                  | $x[n] * h[n] \leftrightarrow X(e^{j\omega})H(e^{j\omega})$                                                                       |
| Multiplication               | $x[n]y[n] \leftrightarrow \frac{1}{2\pi} \int_{2\pi} X(e^{j\theta})Y(e^{j(\omega - \theta)})d\theta$                             |
| Differencing in time         | $x[n] - x[n - 1] \leftrightarrow (1 - e^{-j\omega})X(e^{j\omega})$                                                               |
| Accumulation                 | $\sum_{m=-\infty}^n x[m] \leftrightarrow \frac{X(e^{j\omega})}{1 - e^{-j\omega}} + \pi X(e^{j0}) \sum_k \delta(\omega - 2\pi k)$ |
| Differentiation in frequency | $n x[n] \leftrightarrow j dX(e^{j\omega})/d\omega$                                                                               |
| Real sequence                | $X(e^{-j\omega}) = X^*(e^{j\omega});  X  \text{ even, } \angle X \text{ odd}$                                                    |
| Real and even                | $X(e^{j\omega}) \text{ real and even}$                                                                                           |
| Real and odd                 | $X(e^{j\omega}) \text{ purely imaginary and odd}$                                                                                |
| Even-odd parts               | $\mathcal{E}\{x\} \leftrightarrow \text{Re}\{X\}, \mathcal{O}\{x\} \leftrightarrow j \text{Im}\{X\}$                             |
| Parseval                     | $\sum_n  x[n] ^2 = \frac{1}{2\pi} \int_{2\pi}  X(e^{j\omega}) ^2 d\omega$                                                        |

Discrete-time Fourier transform pairs.

| SEQUENCE                                         | TRANSFORM                                                                                        |
|--------------------------------------------------|--------------------------------------------------------------------------------------------------|
| $\delta[n]$                                      | 1                                                                                                |
| $\delta[n - n_0]$                                | $e^{-j\omega n_0}$                                                                               |
| $u[n]$                                           | $\frac{1}{1-e^{-j\omega}} + \pi \sum_k \delta(\omega - 2\pi k)$                                  |
| $a^n u[n],  a  < 1$                              | $\frac{1}{1-ae^{-j\omega}}$                                                                      |
| $(n+1)a^n u[n],  a  < 1$                         | $\frac{1}{(1-ae^{-j\omega})^2}$                                                                  |
| $\frac{(n+r-1)!}{n!(r-1)!} a^n u[n],  a  < 1$    | $\frac{1}{(1-ae^{-j\omega})^r}$                                                                  |
| $a^{ n },  a  < 1$                               | $\frac{1-a^2}{1-2a \cos \omega + a^2}$                                                           |
| 1 on $ n  \leq N_1$ , 0 beyond                   | $\frac{\sin(\omega(N_1 + \frac{1}{2}))}{\sin(\omega/2)}$                                         |
| $\frac{\sin(Wn)}{\pi n}, 0 < W < \pi$            | 1 on $ \omega  \leq W$ , 0 on $W <  \omega  \leq \pi$                                            |
| 1                                                | $2\pi \sum_k \delta(\omega - 2\pi k)$                                                            |
| $e^{j\omega_0 n}$                                | $2\pi \sum_k \delta(\omega - \omega_0 - 2\pi k)$                                                 |
| $\cos \omega_0 n$                                | $\pi \sum_k [\delta(\omega - \omega_0 - 2\pi k) + \delta(\omega + \omega_0 - 2\pi k)]$           |
| $\sin \omega_0 n$                                | $\frac{\pi}{j} \sum_k [\delta(\omega - \omega_0 - 2\pi k) - \delta(\omega + \omega_0 - 2\pi k)]$ |
| $\sum_{k=\langle N \rangle} a_k e^{jk(2\pi/N)n}$ | $2\pi \sum_k a_k \delta(\omega - \frac{2\pi k}{N})$                                              |
| $\sum_k \delta[n - kN]$                          | $\frac{2\pi}{N} \sum_k \delta(\omega - \frac{2\pi k}{N})$                                        |

Every transform in this section is periodic in  $\omega$  with period  $2\pi$  because the time index is discrete. A periodic sequence has discrete spectral coefficients, while an aperiodic sequence has a continuous spectrum. In each case, discreteness in one domain produces periodicity in the other.

## A.10 Symbols

Symbols and their meanings.

| SYMBOL                 | MEANING                                            |
|------------------------|----------------------------------------------------|
| $x(t), x[n]$           | input signal, continuous and discrete time         |
| $y(t), y[n]$           | output signal                                      |
| $h(t), h[n]$           | impulse response of a linear time-invariant system |
| $\delta(t), \delta[n]$ | unit impulse                                       |
| $u(t), u[n]$           | unit step                                          |
| $E_\infty, P_\infty$   | total energy and average power over all time       |
| $T_0, N_0$             | fundamental period                                 |
| $\omega_0$             | fundamental angular frequency, rad/s or rad/sample |
| $*$                    | convolution                                        |
| $j$                    | imaginary unit, $j^2 = -1$                         |

## PART

## 3

# Notation

Every symbol the course defines, with the chapter that defines it. A symbol is never reused for a second meaning.

| SYMBOL                                           | MEANING                                                                                                                 |
|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| $x(t)$                                           | Continuous-time signal or system input; $t \in \mathbb{R}$ .                                                            |
| $x[n]$                                           | Discrete-time signal or system input; $n \in \mathbb{Z}$ (integer time index).                                          |
| $y(t), y[n]$                                     | System output in continuous and discrete time.                                                                          |
| $h(t), h[n]$                                     | Impulse response: the output of an LTI system when the input is a unit impulse.                                         |
| $\delta(t)$                                      | Continuous-time unit impulse (Dirac delta). It is not an ordinary function. It is defined by its sifting action.        |
| $\delta[n]$                                      | Discrete-time unit impulse: 1 at $n = 0$ , zero elsewhere. An ordinary sequence.                                        |
| $u(t)$                                           | Continuous-time unit step: 1 for $t \geq 0$ , 0 otherwise.                                                              |
| $u[n]$                                           | Discrete-time unit step: 1 for $n \geq 0$ , 0 otherwise.                                                                |
| $E_\infty$                                       | Total energy over an infinite interval ( $R = 1$ normalisation).                                                        |
| $P_\infty$                                       | Time-averaged power over an infinite interval.                                                                          |
| $T_0$                                            | Fundamental period of a continuous-time periodic signal: the smallest $T > 0$ with $x(t) = x(t + T)$ .                  |
| $N_0$                                            | Fundamental period of a discrete-time periodic signal: the smallest integer $N > 0$ with $x[n] = x[n + N]$ .            |
| $\omega_0$                                       | Fundamental angular frequency: $2\pi/T_0$ (rad/s) or $2\pi/N_0$ (rad/sample).                                           |
| $*$                                              | Convolution operator. Defined only for linear time-invariant systems.                                                   |
| $C, a, \alpha, \beta$                            | Complex exponential parameters: $x(t) = Ce^{at}$ , $x[n] = Ce^{\beta n} = C\alpha^n$ with $\alpha = e^\beta$ .          |
| $\mathcal{E}\{ \cdot \}, \mathcal{O}\{ \cdot \}$ | Even and odd parts: $\mathcal{E}\{x\} = \frac{1}{2}[x(t) + x(-t)]$ and $\mathcal{O}\{x\} = \frac{1}{2}[x(t) - x(-t)]$ . |
| $S$                                              | System operator mapping an input signal to an output signal.                                                            |



## Signals and Systems · Formula and Notation Reference

| VERSION | DATE              | CHANGES        |
|---------|-------------------|----------------|
| v0      | 23 September 2026 | First edition. |

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